



**KOÇ
UNIVERSITY**

Text-Based Adventure with LLMs

COMP 491 — Progress Meeting 4

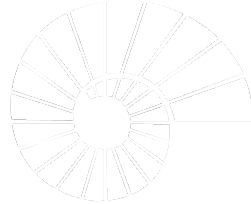
May 8, 2026

Comp491 Project Progress

Project Title: Text-Based Adventure with LLMs

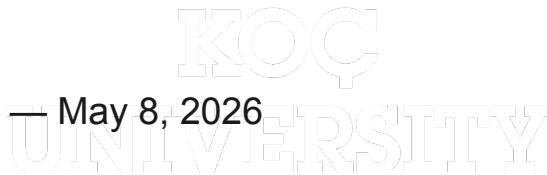
Team Members:

- Kadir Yigit Ozcelik (79975)
- Serdar Yengil (80232)
- Batuhan Karaman (79791)
- Ata Berke Goktekin (80277)



Advisor: Baris Akgun

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Previous Progress Meeting

Progress Meeting 3 Summary (April 3, 2026):

What was accomplished in WP3+WP4:

- Socket.IO multiplayer with room codes, lobby, action batching
- Firebase Auth (Google sign-in) and server-side GameState tracking
- Structured directive validation (MOVE, PICKUP, UNLOCK)
- gpt-image-1.5 scene generation with per-scenario style prompts
- Accusation mechanic with evidence chain validation
- Backend on Google Cloud Run, frontend on Firebase Hosting
- Firestore session persistence

What was planned for WP5:

- Game architecture pivot to host-prompt procedural worlds
- Multiplayer game-over flow, turn limits, finale cinematic
- Voice chat, evidence/notes panel, in-game tutorial
- Spectator mode and replay system
- Procedural narrative depth (red herrings, alibis, NPC backstories)
- Operational hardening (timeouts, usage logging)

Current Progress Meeting

WP5 — Procedural Worlds, Voice Chat, Spectator Mode, Narrative Depth

- The game pivoted from six hardcoded scenarios to a fully procedural pipeline driven by a host prompt; the codebase ships demo-ready on production.

Velvet Shadow v2 — the new game model:

- The host types a theme; the LLM generates the world (rooms, NPCs, items, the murder, the solution) in roughly thirty seconds.
- Single-player removed from the UI; multiplayer-only, two to ten detectives per session.
- Opening and finale are cinematic — AI image, Turkish TTS narration, ambient score, synced typewriter reveal.

Kadir — Frontend, audio, voice and onboarding:

- Three-layer ambient soundtrack and global UI click sound, with a settings popover for separate music and SFX volume.
- Post-game crime-scene reconstruction modal — six to eight chronological beats walking the team through what really happened.
- Push-to-talk voice chat over WebRTC mesh, plus a shared Evidence Board notepad that replaces the old chat tabs.
- Five-step in-game tutorial overlay shown automatically on first play, replayable from the settings menu.

Serdar — Spectators, replay and backend hardening:

- New spectator role at the socket layer — read-only access to a live game, with all write handlers rejecting their input.
- Replay page for finished sessions: scrubbable timeline, play / pause and 1x / 2x / 5x speed, ending with the reconstruction reveal.
- Zod payload validation on every client-emitted event, plus an integration test that verifies Firestore persistence survives a restart.

Ata — Procedural narrative depth and genre awareness:

- Every NPC now ships with an alibi object and a short backstory; the culprit's alibi must contain a discoverable inconsistency.
- Each world contains at least one red herring item that the narrator presents with the weight of real evidence.
- Genre detection on the host prompt injects a tonal style guide — noir, sci-fi, gothic-horror, cyberpunk, period drama.

Batuhan — The pivot itself, AI integration and production deploy (see slide 5)

Batuhan Karaman — Infrastructure & GCP

Completed (Architecture pivot, AI optimization, production deploy):

- Led the Velvet Shadow v2 pivot — replaced six hardcoded scenarios with an LLM-generated world pipeline (strict Zod schema, structured outputs, an automatic repair pass and a hardcoded fallback world) so the host types a theme and the game produces its own rooms, NPCs, evidence and culprit.
- Removed roughly two hundred lines of regex movement and pickup heuristics and collapsed the entire directive surface to a single MOVE — inventory, evidence, sanity and NPC mood now live in narrator prose, giving the model full authorial control.
- Cut the pre-game wait from sixty seconds to thirty by running opening-image generation in parallel with world-text generation, and resolved the resulting race so an early image is baked straight into the ready event instead of being dropped by an old client filter.
- Rewrote the TTS pipeline — moved off broken WAV streams that some browsers refused to decode, switched to MP3, swapped to the warmer 'ash' voice and pre-fetched the audio at story-ready, so the opening cinematic plays at zero latency.
- Shipped real-time multiplayer broadcasts: players in the same room now see each other's actions and the narrator stream live, a session-wide queue banner shows who is currently being written to, and a fifteen-second disconnect grace period keeps reloads and tab throttling from flickering players off the roster.
- Ran today's full production deploy — backend on Cloud Run in europe-west1, web on Cloud Run in us-central1, environment baked at build time, Firebase Hosting CDN auto-invalidated, and the live site verified at velvet-shadow.web.app.

Next Up:

- Add OpenAI-client timeouts, exponential-backoff retry and a fallback model cascade so a single 5xx no longer terminates a live session.
- Add per-IP and per-session rate limiting plus usage and cost telemetry to Cloud Logging so a runaway client cannot blow up the bill.
- Integrate Sentry and a small admin observability dashboard to surface which prompts hit 429s and which sessions end unexpectedly.

Kadir Yigit Ozcelik — Frontend & UI

Completed (Issues #24, #36, #48, #49):

- Built three independent ambient audio layers — lobby, in-game and cinematic — with cross-fade ducking, plus a settings popover that gives separate volume sliders and mute toggles for music and UI sound effects, with preferences persisted to localStorage.
- Added a global UI click sound that hooks every button through document-level event delegation, so the audio touch is consistent across the whole app.
- Built the post-game crime-scene reconstruction modal that opens after the finale and walks the player through six to eight chronological beats; strict ID enums keep the AI from inventing rooms or NPCs, and a single-flight cache prevents duplicate generations across browsers.
- Split reconstruction generation into two model calls — events on a reasoning model, conclusion on a fast non-reasoning one — after the original single call kept truncating the closing paragraph against the same token budget.
- Shipped V-key push-to-talk voice chat over WebRTC mesh with Socket.IO acting as a signaling-only relay; holding V transmits to all teammates, typing 'v' inside any input field passes through cleanly, and the speaker's avatar lights up live.
- Replaced the old room and direct chat tabs with a single shared Evidence Board notepad — every note broadcasts to all players, treated as out-of-game thinking material that the narrator never reads.
- Built a five-step in-game tutorial overlay that triggers on first play and highlights the turn counter, chat area, action input, accuse button and settings; keyboard navigable and replayable from the settings menu.

Next Up:

- Polish the mobile and small-screen layout — header buttons, the accusation banner and the action input still need responsive media queries.
- Run end-to-end voice-chat verification against the production HTTPS deployment, where getUserMedia and STUN traversal behave differently than on localhost.
- Add an in-app changelog panel for the demo audience so newcomers can see what shipped in the latest sprint.

Serdar Yengil — Backend & API

Completed (Issues #52, #53):

- Added a spectator role to the Socket.IO handshake so observers receive the full session state and the live narrator stream while every write handler rejects them — invited guests can watch a real game without being able to interfere.
- Built the spectator UI on the home page and adapted the in-game view for it: chat input, accuse and leave are hidden and a Spectator badge appears in the header.
- Built the replay system as a dedicated page with a scrubbable timeline, play / pause and 1x / 2x / 5x speed controls, ending with the same reconstruction modal as the live game.
- Added Zod schema validation to every client-emitted Socket.IO event so malformed or hostile payloads never reach a handler — the server returns an INVALID_PAYLOAD error event instead.
- Wrote a Firestore integration test that creates a session, populates it, syncs, simulates a server restart, rehydrates the store and asserts byte-for-byte equality of the recovered state.

Next Up:

- Add session-resume on the spectator side so a dropped observer reconnects and catches up automatically without a manual refresh.
- Profile and tune the action-batching window under realistic four-player load, since the current values were chosen against solo tests.
- Add a server-side per-session rate limit on action submission to complement the per-IP limit at the OpenAI call layer.

Ata Berke Goktekin — Game Design & Prompts

Completed (Issues #54, #55):

- Extended the procedural world schema with three new fields — every NPC now ships with an alibi object (claimed location, claimed activity, optional corroboration and inconsistency) and a short backstory the narrator can pull from.
- Made every world contain at least one red-herring item, and constrained the generator so the culprit's alibi must hide a discoverable inconsistency while innocent NPCs stay consistent but information-limited.
- Updated the narrator prompt to use the new fields so when a player questions a suspect the narrator delivers their alibi naturally, draws on the backstory for voice and gives red herrings the same narrative weight as real evidence.
- Updated the post-game reconstruction so the conclusion explicitly acknowledges any red herring ("the broken letter was a misdirection"), letting players see which leads were intentional dead ends.
- Built a genre-aware style amplifier — a keyword heuristic with a small LLM fallback detects the genre of the host prompt and injects a tonal style guide into both the world generator and the narrator's system prompt.
- Authored concrete style guides for noir, sci-fi, gothic-horror, cyberpunk and period drama, plus a generic fallback — the same procedural pipeline now produces tonally distinct worlds across themes.

Next Up:

- Broaden the genre library with fantasy, modern thriller and cosmic horror, and tighten the noir guide based on playtest feedback.
- Add a pre-game world-quality check where the LLM critiques its own world and re-rolls if it spots unreachable rooms, missing inconsistencies or no red herring.
- Run a five-prompt diversity playtest to measure whether the new procedural depth genuinely shifts gameplay between themes.

References

- [1] N. Montfort, *Twisty Little Passages: An Approach to Interactive Fiction*. MIT Press, 2003.
- [2] A. Plotkin, *The Inform Designer's Handbook*, 2004.
- [3] P. Ammanabrolu et al., *How to Avoid Being Eaten by a Grue*, arXiv:2006.07409, 2020.
- [4] N. Walton, *AI Dungeon*, Latitude, 2019. <https://aidungeon.io>
- [5] OpenAI, *GPT-5 API Documentation*, 2025. <https://platform.openai.com/docs>
- [6] N. Shaker et al., *Procedural Content Generation in Games*. Springer, 2016.
- [7] Socket.IO Documentation, 2024. <https://socket.io/docs/v4/>
- [8] WebRTC API, MDN Web Docs, 2024. https://developer.mozilla.org/en-US/docs/Web/API/WebRTC_API

Attachment

Live Application:

- <https://velvet-shadow.web.app>
- Backend: Cloud Run (europe-west1)
- Database: Firestore (comp491-akboys-2026)

GitHub Repository:

- <https://github.com/bakaraman/comp491-akboys>
- 8 issues closed this sprint, 2 PRs merged + 1 branch direct-merged
- 2 open issues for WP6 (Batuhan's operational hardening — in progress)

